IT SUPPORT TROUBLESHOOTING GUIDE

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Entry-Level IT Support Portfolio Project

March 2026

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Introduction

This guide documents common IT support issues and structured troubleshooting approaches used to diagnose and resolve technical problems.

Each issue is presented in the following format:

- Problem.
- Possible Causes.
- Step-by-Step Solution.
- Commands Used (if any).
- Why This Works.

This demonstrates practical diagnostic thinking and structured IT support methodology.

Windows Operating System Issues

Issue 1: No Internet Access (Windows)

Problem:

Computer shows “No Internet Access” despite being connected to WiFi.

Possible Causes:

- Router malfunction.
- Incorrect IP configuration.
- DNS resolution failure.
- Disabled network adapter.

Step-by-Step Solution:

1. Restart router and PC.
2. Open Command Prompt as Administrator.
3. Run:

   ```
   ipconfig /release
   ipconfig /renew
   ```

4. Check IP address:

   ```
   ipconfig
   ```

5. Test connectivity:

   ```
   ping 8.8.8.8
   ```

Commands Used:

- `ipconfig`
- `ping`

Why This Works:

Releasing and renewing refreshes IP assignment from DHCP server. Pinging verifies network communication with an external server.

Issue 2: Slow Computer Performance

Problem:

Computer takes long to boot and programs freeze.

Possible Causes:

- Too many startup programs.
- Low RAM.
- Malware.
- Disk fragmentation.

Step-by-Step Solution:

1. Open Task Manager - Startup tab.
2. Disable unnecessary startup programs.
3. Run Disk Cleanup.
4. Run Windows Defender scan.
5. Check RAM usage in Task Manager.

Why This Works:

Reducing startup load improves boot speed and frees system memory.

Printer Troubleshooting

Issue 3: Printer Not Printing

Problem:

Printer is connected but not printing documents.

Possible Causes:

- Printer set to offline.
- Print spooler service stopped.
- Outdated drivers.
- Paper jam.

Step-by-Step Solution:

1. Check if printer is set as default.
2. Open Services - Restart “Print Spooler”.
3. Remove printer and reinstall driver.
4. Check physical connection and paper tray.

Why This Works:

Restarting spooler clears stuck print jobs and resets communication between system and printer.

Issue 4: Printer Showing Offline

Step-by-Step Solution:

1. Open Control Panel - Devices and Printers.
2. Right-click printer - Uncheck “Use Printer Offline”.
3. Restart printer and PC.

Why This Works:

Forces system to re-establish communication with printer device.

Network Connectivity Troubleshooting

Issue 5: Website Not Loading

Problem:

WiFi connected but websites not opening.

Possible Causes:

- Domain Name System (DNS) issue.
- Firewall blocking access.
- Browser problem.

Step-by-Step Solution:

1. Test network:

   ```
   ping 8.8.8.8
   ```

2. Flush DNS:

   ```
   ipconfig /flushdns
   ```

3. Change DNS to:

   - `8.8.8.8`
   - `8.8.4.4`

Why This Works:

Flushing clears corrupted DNS cache. Changing DNS ensures reliable domain resolution.

DNS & IP Configuration Diagnostics

Useful Network Commands

a) `ipconfig`
   Displays IP configuration details.
b) `ipconfig /all`
   Shows detailed adapter information.
c) `ping`
   Tests connectivity between two devices.
d) `tracert`
   Shows packet path to destination.
e) `nslookup`
   Tests DNS resolution.

These tools help isolate whether a problem is local, router-based, or external.

Microsoft Office Issues

Issue 6: Microsoft Word Crashes on Startup

Possible Causes:

- Corrupt add-ins.
- Damaged installation.
- Outdated version.

Step-by-Step Solution:

1. Open Run - Type:

   ```
   winword /safe
   ```

2. Disable add-ins.
3. Go to Control Panel - Repair Office.
4. Update Microsoft Office.

Why This Works:

Safe Mode isolates faulty extensions causing crash.

Issue 7: Activation Error

Step-by-Step Solution:

1. Check internet connection.
2. Sign in with correct Microsoft account.
3. Run Office Activation Troubleshooter.

Why This Works:

Activation requires valid license verification with Microsoft servers.

Basic Hardware Troubleshooting

Issue 8: Computer Not Powering On

Possible Causes:

- Faulty power cable.
- Dead battery.
- PSU failure.

Step-by-Step Solution:

1. Check power cable.
2. Test with another outlet.
3. Remove battery (if laptop) and test with charger.
4. Listen for fan or LED indicators.

Why This Works:

Isolates power delivery issues before assuming motherboard failure.

Conclusion

This troubleshooting guide demonstrates structured diagnostic thinking in IT Support.

By following systematic steps:

- Identify the problem.
- Analyze possible causes.
- Test logically.
- Confirm resolution,

IT professionals can efficiently resolve technical issues while minimizing downtime.